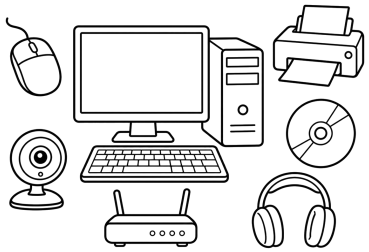
TEACHER GUIDE

ANSWERS AND MARKS

GRADE 10 · INFORMATION AND COMMUNICATION TECHNOLOGY

UNIT 2
Fundamentals of a Computer System



! FOR THE TEACHER ONLY

This booklet has the answers to every exercise in the Unit 2 Student Workbook, in the same order. The exercise letters match the workbook.

i WHAT IS IN THIS GUIDE

- **Marks and pacing** - every exercise and its marks
- **Class marks record** - one page to copy for each class
- **Practicals at a glance** - what to check, safety, a quiz, a practical test
- **Answers** - chapters 1 to 4, in the same order as the workbook
- **Unit 2 Test** - the marking scheme and the common mistakes
- **Extra** - textbook links, homework ideas and teaching tips

★ BEFORE THE FIRST LESSON

- ☐ Old devices to show: a keyboard, a mouse, a hard disk, discs, a flash drive.
- ☐ A discarded hard disk is opened (by you) for Practical 1.
- ☐ You know where the lab switch and router are, and how to show the firewall.

"Students remember the devices they hold in their hands. Bring the hardware to the class."

- THE KING OF ICT

TEACHER GUIDE

Marks and Pacing

You decide how much class time each part needs

Chapter	Exercise	Marks
Chapter 1	A Fill in the blanks	8
	B Analog, digital or hybrid?	6
	C True or false	6
	D Match the computers	6
	E Complete Figure 2.10	7
Chapter 2	A Complete the printer tree	5
	B Input, output or both?	8
	C Which device would you use?	8
	D Impact or non impact?	6
	E Label the keyboard	5
	F Match them	6
	G Choose the correct answer	6
Chapter 3	A Magnetic, optical or solid state?	8
	B Label the hard disk	7
	C RAM, ROM or cache?	6
	D Name the ports	8
	E Which port would you use?	6
	F Complete the tables	6
Chapter 4	A Match the network devices	6
	B Crossword	10
	C Simplex, half-duplex or full duplex?	6
	D Complete the media tree	8
	E LAN, PAN, MAN or WAN?	6
	F Name the topologies	5
	G Benefit or disadvantage?	6
	H Answer these	4
End	Unit 2 Test	40
TOTAL		**209**

SUGGESTED SHAPE OF ONE CHAPTER

READ	Teach the chapter pages. Students read with you and underline the key words.
DO	The practical tasks on real computers, in pairs.
WRITE	The written exercises. Walk around and help.
MARK	Mark the work together. Fill the tick boxes.

TIP: Before Practical 2, shut down the computers and switch them off at the mains.

TEACHER GUIDE

Class Marks Record

Copy this page for each class

No	Student name	Ch 1 (33)	Ch 2 (44)	Ch 3 (41)	Ch 4 (51)	Test (40)	Total (209)
1							
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TEACHER GUIDE

Practicals at a Glance

The textbook activities on pages 55 - 68

Practical	Students do	Check that ...
1 (p. 55)	Advantages and disadvantages of magnetic, optical and solid state media; the parts of an opened hard disk	the platter, spindle, head, actuator arm, actuator and both connectors are found; nobody touches the platter.
2 (p. 58)	Observe and identify the ports of a lab computer	the computer is switched off at the mains; every port is named with its use; the back panel is drawn.
3 (p. 62)	Copper wire or optical fibre? A discussion of speed, distance and price	fibre: fastest, long distances, expensive; copper (UTP): cheap and flexible, up to about 100 m.
4 (p. 65)	Observe the firewall of the operating system	nothing is changed; students can say what a firewall does.

! SAFETY AND MATERIALS

- Old devices to show: a keyboard, a mouse, a hard disk, CDs / DVDs, a flash drive, a memory card, cables.
- Practical 1: open the discarded hard disk yourself before the lesson. Keep the magnets away from phones.
- Practical 2: the computers are shut down and switched off at the mains before students look at the ports.
- Practical 4: use a teacher account; students look at the firewall settings but change nothing.

Textbook links

Workbook	Textbook pages	Activities
Chapter 1 - The computer and its classification	37 - 41	-
Chapter 2 - Input devices, output devices and the CPU	42 - 50	-
Chapter 3 - Memory, storage and ports	51 - 58	55, 58
Chapter 4 - Data communication and computer networks	58 - 71	62, 65, 68
Unit 2 in One Page	72 (summary)	-

★ TEACHING TIPS

- Bring real devices to the class: an old keyboard, a mouse, a hard disk, a CD, a flash drive and a network cable.
- Draw Figure 2.10 on the board step by step. Ask students to follow one piece of data from the keyboard to the printer.
- Ask students to look at their phone chargers at home: which USB type do they use?

? QUICK ORAL QUIZ - START OR END A LESSON WITH IT

Ask the class ...	Answer
1. Name the five features of a computer.	**Speed, accuracy, efficiency, versatility, storing**
2. Which computer has the highest computing power?	**Super computer**
3. Give an example of a hybrid computer.	**The ECG machine**
4. Which device reads the numbers on a cheque?	**MICR**
5. Is a touch screen input or output?	**Both**
6. Which printer uses toner?	**Laser printer**
7. Which memory keeps the BIOS?	**ROM**
8. How many holes has a VGA port?	**15**
9. Is a telephone call simplex, half or full duplex?	**Full duplex**
10. Which device is also called a gateway?	**Router**

> PRACTICAL TEST - 30 MINUTES, 20 MARKS (OPTIONAL)

In the lab, one student at a time. The computer is switched off for steps 2 and 3.

Step	The student must ...	Marks
1	Name the input, output and storage devices on your table.	4
2	Point to and name five ports at the back of the system unit, with their uses.	5
3	Connect a keyboard, a mouse and a monitor to the right ports (the teacher checks).	3
4	Find the processor speed (GHz) and the RAM size of the computer (This PC - Properties).	4
5	Show that the firewall is on (Control Panel - System and Security).	2
6	Name the topology of the lab network and the device in its centre.	2
	Total	**20**

★ **TIP:** Say it every lesson: dotted arrows carry data and instructions; solid arrows carry control signals.

ANSWERS

Chapter 1

The computer and its classification

A Fill in the blanks

8 marks

1. processes	**2.** million	**3.** versatility	**4.** terminals
5. server	**6.** micro	**7.** temperature	**8.** Control Unit

B Analog, digital or hybrid?

6 marks

The computer	Analog	Digital	Hybrid
A speedometer in a vehicle	☑	☐	☐
An ECG machine in a hospital	☐	☐	☑
The laptop you use at home	☐	☑	☐
A meteorological (weather) machine	☑	☐	☐
A road lamp that switches on with a sensor	☑	☐	☐
A smart phone	☐	☑	☐

C True or false

6 marks

1. **T - A computer gives correct output only when correct instructions and data are given.**
2. **F - Super computers are small, cheap and common.**
3. **F - Mini computers are more powerful than mainframe computers.**
4. **T - Mainframe and mini computers are now called server computers.**
5. **F - In Figure 2.10, control signals are shown by a dotted line.**
6. **T - Data from an input device first goes to the primary memory.**

D Match the computers

6 marks

1. Super computer → **C**	**2.** Mainframe computer → **F**	**3.** Mini computer → **A**
4. Micro computer → **D**	**5.** Analog computer → **E**	**6.** Hybrid computer → **B**

E Complete Figure 2.10

7 marks

1. Input devices	**2.** Main memory	**3.** Secondary memory	**4.** Output devices
5. Registers	**6.** ALU (Arithmetic and Logical Unit)	**7.** Control Unit (CU)	

1 mark for each box. Accept "primary memory" for box 2.

★ TEACHING TIPS

- Biggest to smallest: Super, Mainframe, Mini, Micro. Let students make their own memory sentence.
- Draw an analog wave and a digital (square) wave. Ask which one the needle of a speedometer shows.
- Let pairs sort device cards into input, output and both. The touch screen always starts a good discussion.

ANSWERS

Chapter 2

Input devices, output devices and the CPU

A Complete the printer tree

5 marks

1. Impact printers	**2.** Non impact printers	**3.** Line printer	**4.** Laser printer	**5.** Thermal printer

1 mark each. Boxes 4 and 5 may be in either order.

B Input, output or both?

8 marks

The device	Input	Output	Both
Light pen	☑	☐	☐
Plotter	☐	☑	☐
Touch screen	☐	☐	☑
MICR	☑	☐	☐
Multimedia projector	☐	☑	☐
Webcam	☑	☐	☐
Thermal printer	☐	☑	☐
OMR device	☑	☐	☐

C Which device would you use?

8 marks

1. MICR	**2.** OMR device	**3.** OCR device	**4.** Barcode reader
5. Light pen	**6.** Plotter	**7.** Thermal printer	**8.** Multimedia projector

1 mark each. Accept "OMR" and "OCR" without the word "device".

D Impact or non impact?

6 marks

The printer ...	Impact	Non impact
A head or a needle strikes an ink ribbon.	☑	☐
It is not noisy.	☐	☑
Its ribbon is cheap, so banks and shops use it for receipts.	☑	☐
Toner (dried ink) from a cartridge is put on the paper.	☐	☑
It prints about 3000 lines per minute.	☑	☐
Heat is used to print the document.	☐	☑

E Label the keyboard

5 marks

1. Function keys	**2.** Typing keys	**3.** Special keys	**4.** Control keys	**5.** Numeric keys

Special keys are keys such as Esc, Tab, Caps Lock, Shift, Enter and Print Screen; control keys are Ctrl, Alt, the Windows key and the arrow and page keys.

F Match them

6 marks

1. CRT monitor → **B**	**2.** LCD monitor → **D**	**3.** LED screen → **F**
4. ALU → **A**	**5.** Control Unit → **E**	**6.** Memory registers → **C**

G Choose the correct answer

6 marks

1. (b) MHz and GHz	**2.** (a) a soft copy
3. (c) Touch screen	**4.** (c) brain
5. (b) Line printer	**6.** (c) 101 to 108 keys

ANSWERS

Chapter 3

Memory, storage and ports

A Magnetic, optical or solid state?

8 marks

The device	Magnetic	Optical	Solid state
Hard disk	☑	☐	☐
CD-RW	☐	☑	☐
USB flash drive	☐	☐	☑
Magnetic tape	☑	☐	☐
DVD	☐	☑	☐
Memory card (SD card)	☐	☐	☑
Blu-ray disc	☐	☑	☐
Floppy disk	☑	☐	☐

B Label the hard disk

7 marks

1. Platter	**2.** Spindle	**3.** Head	**4.** Actuator arm
5. Power connector	**6.** IDE connector	**7.** Actuator	

C RAM, ROM or cache?

6 marks

The memory ...	RAM	ROM	Cache
Its data is lost when the power is switched off.	☑	☐	☐
It keeps the BIOS (the commands to boot).	☐	☑	☐
It is between the CPU and the primary memory.	☐	☐	☑
It is often called the main memory.	☑	☐	☐
It is small but fast; also called CPU memory.	☐	☐	☑
It is non-volatile and fixed to the motherboard.	☐	☑	☐

D Name the ports

8 marks

1. HDMI port	**2.** PS/2 port	**3.** VGA video port	**4.** Audio ports / jacks
5. Ethernet port (RJ45)	**6.** Serial port	**7.** USB port	**8.** Parallel port

1 mark each. Accept "graphic port" for VGA and "network port" for the Ethernet port.

E Which port would you use?

6 marks

1. HDMI port (or VGA port)	**2.** Ethernet port (RJ45)	**3.** Audio port (pink jack)
4. Parallel port	**5.** Serial port	**6.** PS/2 port (green)

F Complete the tables

6 marks

1. CD: **650 - 900 MB**	**2.** DVD: **4.7 - 9.4 GB**	**3.** Blu-ray disc: **25 - 128 GB**
4. CD-ROM: **read only (you can only read it)**	**5.** DVD-R: **can be written once**	**6.** CD-RW: **can be erased and re-written**

1 mark each (textbook page 54).

ANSWERS

Chapter 4

Data communication and computer networks

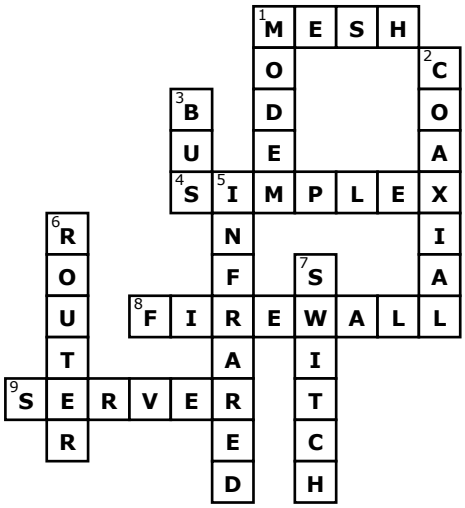
A Match the network devices

6 marks

1. NIC → **C**	**2.** Hub → **E**	**3.** Switch → **B**
4. Router → **F**	**5.** Modem → **A**	**6.** Firewall → **D**

B Crossword

10 marks



Across: 1 MESH, 4 SIMPLEX, 8 FIREWALL, 9 SERVER

Down: 1 MODEM, 2 COAXIAL, 3 BUS, 5 INFRARED, 6 ROUTER, 7 SWITCH

C Simplex, half-duplex or full duplex?

6 marks

The communication	Simplex	Half-duplex	Full duplex
Watching television	☑	☐	☐
Talking on the telephone	☐	☐	☑
A walkie-talkie used by security forces	☐	☑	☐
Listening to the radio	☑	☐	☐
Browsing the Internet to get information	☐	☑	☐
Getting a printout	☑	☐	☐

D Complete the media tree

8 marks

1. Guided / wired	**2.** Unguided / wireless	**3.** Coaxial cable	**4.** Fibre optics
5. Radio waves	**6.** Microwaves	**7.** Infrared	**8.** UTP

1 mark each. Boxes 3 and 4 may be in either order, and so may boxes 5, 6 and 7.

E LAN, PAN, MAN or WAN?

6 marks

The network	LAN	PAN	MAN	WAN
The computers of your school lab	☑	☐	☐	☐
The Internet	☐	☐	☐	☑
The branches of a bank in one city, joined together	☐	☐	☑	☐
A phone joined to earphones with Bluetooth	☐	☑	☐	☐
The computers of one office building	☑	☐	☐	☐
A university and its branches in a city	☐	☐	☑	☐

F Name the topologies

5 marks

1. Ring topology	**2.** Star topology	**3.** Mesh topology	**4.** Bus topology	**5.** Tree topology

G Benefit or disadvantage?

6 marks

Networking ...	Benefit	Disadvantage
One printer can be shared by many users.	☑	☐
A virus can spread to the other computers.	☐	☑
Hackers may try to get into the server.	☐	☑
Software can be installed on a server and shared.	☑	☐
A trained network administrator is needed.	☐	☑
Users can reach their documents any time, anywhere.	☑	☐

H Answer these

4 marks

1. Write two differences between a hub and a switch. (2 marks)

A hub sends the data to all the computers; a switch sends it only to the relevant computer. A hub uses half duplex mode; a switch uses full duplex mode (a hub can congest the network). 1 mark each.

2. What is a Personal Area Network (PAN)? Give one example. (2 marks)

The smallest network: it joins devices within a short range (about 1 to 10 metres), often with Bluetooth, Wi-Fi, infrared or USB (1 mark). For example a mobile phone joined to earphones or to a laptop (1 mark).

TIP: Explain MICR with a real cheque (cover the account number). Students remember the magnetic numbers.

MARKING SCHEME

Unit 2 Test

Total 40 marks

1 Choose the correct answer

8 marks (1 each)

1. (c) Super	**2.** (b) Plotter
3. (b) RAM	**4.** (a) ALU
5. (c) Ethernet	**6.** (b) half-duplex
7. (c) fibre optics	**8.** (b) A and C only

2 Fill in the blanks

8 marks (1 each)

1. hybrid	**2.** ROM	**3.** cache	**4.** 15
5. hard	**6.** mesh	**7.** router	**8.** radio

3 Look at the pictures

8 marks (1 + 1 each)

1. VGA (video) port - connects a monitor or a multimedia projector	**2. Ethernet port (RJ45)** - joins the computer to a network router or switch
3. Ring topology - data moves in a circle; one break stops the network	**4. Bus topology** - all computers share one main cable, the backbone

1 mark for the name and 1 mark for the use or a correct fact.

4 Answer these

16 marks

1. Name the three main parts of the CPU and write what each part does. (3 marks)

ALU - does the mathematical and logical functions; Control Unit - controls all the devices; memory registers - temporarily store the data for the ALU and its results. 1 mark for each part with its job.

2. Write two differences between RAM and ROM. (2 marks)

RAM is volatile (its data is deleted when the power is off), ROM is non-volatile; RAM can be read and written, ROM is read only and keeps the BIOS. Any two, 1 mark each.

3. Name the three types of secondary storage devices. Give one example of each. (3 marks)

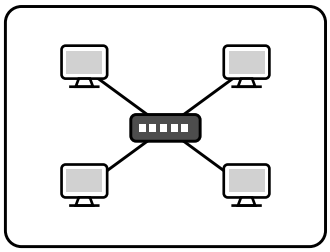
Magnetic media - hard disk / magnetic tape; optical media - CD / DVD / Blu-ray; solid state devices - USB flash drive / memory card. 1 mark for each type with an example.

4. What is the difference between half-duplex mode and full duplex mode? Give one example of each. (2 marks)

Half-duplex: one direction at a time, e.g. a walkie-talkie or browsing the Internet. Full duplex: both directions at the same time, e.g. a telephone conversation. 1 mark for each.

5. Draw a star topology with four computers. Label the device in the centre. (3 marks)

A switch or hub in the centre (1 mark), four computers (1 mark), each joined to the centre by its own cable (1 mark).



6. Write two benefits and one disadvantage of computer networks. (3 marks)

Benefits (any two): share data and information, share resources such as printers, less storage space, central software, access any time, security, e-mail. Disadvantage (any one): security issues, a breakdown can affect others, viruses spread, server breakdown, a trained administrator is needed. 1 mark each.

WHAT STUDENTS USUALLY GET WRONG

- They call the touch screen only an output device. It is an input **and** an output device.
- They mix up MICR (cheques), OCR (the text of a document) and OMR (pen or pencil marks).
- They say that ROM loses its data when the power is off. **RAM** is volatile; ROM keeps the BIOS.
- They write that cache memory is bigger than RAM. It is **smaller but faster**.
- They mix up VGA (15 holes, video), the parallel port (25 holes, printer) and the serial port (modem).
- They think a walkie-talkie is full duplex. Only one person talks at a time: **half-duplex**.
- They say a hub sends data to one computer. That is the **switch**; a hub sends to all.

HOMEWORK AND EXTRA PRACTICE

- Make a table of ten devices at home: input, output or storage, and the port each one uses.
- Textbook Activity, page 55: the advantages and disadvantages of magnetic, optical and solid state media.
- Textbook Activity, page 68: two examples each of a LAN, a MAN and a WAN in Sri Lanka.
- Fast learners: find the processor speed, the RAM and the hard disk size of a lab computer and compare them with a friend's computer at home.

TEXTBOOK NOTES

- The textbook writes VGA as "Video Graphics Adaptor" and MICR as "Magnetic Ink Character Reader". Accept "Video Graphics Array" and "Magnetic Ink Character Recognition" too.
- Two sections of the textbook are numbered 2.5 (ports and data communication). The workbook numbers its sections 2.1 - 2.17 in its own order.

TEACHING TIP

- Show an OMR answer sheet from a past exam. Ask why the circles must be shaded fully with a pencil.